

Optimization and Redistribution Strategies for Italian Renewable Energy Communities

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Abstract—In this paper we consider the optimal operation of an energy community receiving incentives for energy virtually shared among the community members. A similar incentive scheme for renewable energy communities is adopted in Italy since 2020. The operational problem is formulated as the maximization of the profit of the community over a given time horizon. The profit includes the incentive for the virtual self-consumption. The optimization exploits all available sources of flexibility, such as controllable loads and generators, and battery energy storage systems. Compared to previous work of the authors, the mixed-integer formulation proposed in this paper requires a smaller number of continuous variables. Still, the number of binary variables may be prohibitive when the size of the community grows. For this reason, an equivalent formulation in the form of convex maximization is derived, involving only continuous variables. We show examples where the latter problem can be solved in reasonable time, while the solution process of the mixed integer problem gets stuck. Moreover, a simple strategy to fairly redistribute the community benefits among its members is discussed.

Index Terms—Energy community, profit maximization, mixed integer optimization, convex maximization, redistribution strategies.

I. INTRODUCTION

Renewable energy communities (RECs) are aggregations with the goal of providing environmental, economic, and social benefits to their members, or the local areas where they operate, through the sharing of energy from renewable sources [1]. Members may be of several types (e.g., residential, commercial, industrial). They participate by consuming, producing, and/or storing energy. Thanks to Agenda 2030 [2] and the Directive 2019/944 [1], the EU paved the way for energy transition of the member states, suggesting that energy communities may be a key tool to that purpose. Different national and regional laws regulate the creation of energy communities and their incentivization, with the aim of driving citizens to participate. The current Italian regulation assigns an incentive for the self-consumption at the community level, defined in [3] as “the minimum between the energy fed into the network and the energy consumed by the community members” in a given time period, typically set to 1 hour. This incentive scheme, which has been adopted in Italy since 2020

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[4], is aimed at providing system level benefits, by encouraging investments in renewable energy sources and increasing energy efficiency through on-site consumption. On the community side, particular attention should be given to guaranteeing that the incentive for self-consumption is fairly shared among the participants [5]. To this purpose, the community may implement suitable rules for the redistribution of benefits among its members [6], ensuring that all members do not increase their individual bills [7], and no group of users will find advantageous exiting the community.

Regarding the management of energy communities, the optimal allocation of resources, combined with the optimization of energy sharing, achieved by means of the flexibility of consumers and prosumers, and the better control over consumption and generation, may increase the benefits coming from active participation [8]. Apart from incentive schemes, the exchange of energy within a community can be facilitated by the creation of a local energy market designed for this purpose. The local market can be run either in a centralized or decentralized manner, see [10].

A. Paper contribution

In this paper, we consider a community model based on the Italian regulation. All the members (also termed entities) are connected to the main grid, and exchange energy through it. The community does not organize a local energy market. Coupling among the members is established through the incentive scheme based on the virtual self-consumption described above. We assume a centralized organization, where a community operator collects data from all the members, and optimizes the operation of the community taking the incentive for the virtual self-consumption explicitly into account in the economic balance. This coordinated scheme contrasts with the uncoordinated one, where each member optimizes its operation individually, and the incentive is evaluated a posteriori. In our coordinated scheme, the operational problem is formulated as the maximization of the profit of the community over a given time horizon. The profit includes the incentive for the virtual self-consumption. The optimization exploits all available sources of flexibility, such as controllable loads and generators, and battery energy storage systems. The mixed-integer formulation proposed in this paper improves the one in [11], since it requires a smaller number of continuous

TABLE I
NOMENCLATURE

Constants and sets

Δ_T	duration of a time period [h]
U	number of entities forming the community
T	number of time periods
$\mathcal{U}$	set of all the entities, with $\mathcal{U} = \{1, \dots, U\}$
$\mathcal{T}$	set of the time indexes, with $\mathcal{T} = \{1, \dots, T\}$

Parameters

$I_{u,t}^{min}$	lower bound to energy demand of entity u at time t [kWh]
$I_{u,t}^{nom}$	nominal energy demand of entity u at time t [kWh]
$I_{u,t}^{max}$	upper bound to energy demand of entity u at time t [kWh]
$E_{u,t}^{min}$	lower bound to energy production of entity u at time t [kWh]
$E_{u,t}^{nom}$	nominal energy production of entity u at time t [kWh]
$E_{u,t}^{max}$	upper bound to energy production of entity u at time t [kWh]
$S_{u,t}^{min}$	lower bound to battery capacity of entity u [kWh]
$S_{u,t}^{max}$	upper bound to battery capacity of entity u [kWh]
S_u^{ini}	battery capacity of entity u at time $t = 0$ [kWh]
S_u^{end}	battery capacity of entity u at time $t = T$ [kWh]
P_u^{dis}	maximum discharging power of battery of entity u [kW]
P_u^{cha}	maximum charging power of battery of entity u [kW]
η_u^{dis}	discharging efficiency of battery of entity u
η_u^{cha}	charging efficiency of battery of entity u
$E_{u,t}^{cap}$	upper bound to power export of entity u to the grid at time t [kW]
$I_{u,t}^{cap}$	upper bound to power import of entity u from the grid at time t [kW]

$\pi_{u,t}^{egr}$	unit price of energy exported to the grid by entity u at time t [€/kWh]
$\pi_{u,t}^{igr}$	unit price of energy imported from the grid by entity u at time t [€/kWh]
$\pi_{u,t}^{fle}$	unitary cost for activating energy demand flexibility of entity u at time t [€/kWh]
$\pi_{u,t}^{reg}$	unitary cost for regulating energy production of entity u at time t [€/kWh]
$\pi_{u,t}^{sto}$	unitary cost for usage of energy storage of entity u at time t [€/kWh]
π_t^{inc}	unit incentive for virtual self-consumption at time t [€/kWh]

Variables

$i_{u,t}$	energy demand of entity u at time t [kWh]
$e_{u,t}$	energy production of entity u at time t [kWh]
$i_{u,t}^{gr}$	energy imported from the grid by entity u at time t [kWh]
$e_{u,t}^{gr}$	energy exported to the grid by entity u at time t [kWh]
$\delta_{u,t}$	binary variable expressing if entity u at time t is exporting or not
$p_{u,t}^{dis}$	discharging power of battery of entity u at time t [kW]
$p_{u,t}^{cha}$	charging power of battery of entity u at time t [kW]
$s_{u,t}$	battery capacity of entity u at time t [kWh]
$\zeta_t, \epsilon_{u,t}^e, \epsilon_{u,t}^i$	auxiliary variables

variables. Still, the solution of the formulated problem may turn out to be prohibitive in real world problems, due to the number of binary variables which grows proportionally with the size of the community. This is an issue common to mixed integer approaches, and shared by other formulations for the Italian case presented in the literature [11]–[13]. To cope with this issue, an equivalent formulation in the form of convex maximization is derived, involving only continuous variables. We show examples where the latter problem can be solved in reasonable time, while state-of-the-art solvers crash while solving the mixed integer problem. Moreover, a simple strategy to fairly redistribute the community benefits among its members is discussed.

B. Paper organization

The paper is organized as follows. Section II describes the mixed-integer and convex maximization formulations of the community operational problem. A simple redistribution strategy and its properties are presented in Section III, while Section IV reports a toy example, illustrating the redistribution strategy, and an application with real consumption, generation, and price data, showing the computational feasibility of the convex maximization formulation as compared to the mixed-integer one. Finally, Section V draws the conclusions.

II. PROBLEM FORMULATION

We consider energy communities consisting of U entities, each represented by a grid-connected microgrid. For the sake

of simplicity, we assume that each entity may encompass a load, a generator, and/or a battery energy storage system (BESS). However, the model can be easily extended to accommodate entities that comprise multiple loads, generators, and BESSs.

A. MILP formulation

For each entity u of the community, the energy demand $i_{u,t}$ and the energy production $e_{u,t}$ at time t have nominal values $I_{u,t}^{nom}$ and $E_{u,t}^{nom}$, respectively. Deviations from the nominal values are allowed (with a cost specified later), provided that $i_{u,t}$ and $e_{u,t}$ do not exceed specified bounds:

$$I_{u,t}^{min} \leq i_{u,t} \leq I_{u,t}^{max} \quad (1)$$

$$E_{u,t}^{min} \leq e_{u,t} \leq E_{u,t}^{max}. \quad (2)$$

Ad hoc constraints can be added to model the typical energy payback effect of flexible domestic customers [14], e.g.:

$$\sum_{t \in \mathcal{T}} i_{u',t} = \sum_{t \in \mathcal{T}} I_{u',t}^{nom} \quad (3)$$

for all $u' \in \mathcal{D}$, where $\mathcal{D} \subseteq \mathcal{U}$ is the set of entities with domestic loads.

For the BESS of entity u , the charging and discharging power, as well as the battery capacity, are bounded as follows:

$$0 \leq p_{u,t}^{cha} \leq P_u^{cha} \quad (4)$$

$$0 \leq p_{u,t}^{dis} \leq P_u^{dis} \quad (5)$$

$$S_u^{min} \leq s_{u,t} \leq S_u^{max}. \quad (6)$$

The battery dynamics is described by the difference equation:

$$s_{u,t} = s_{u,t-1} + \eta_u^{cha} p_{u,t}^{cha} \Delta T - \frac{1}{\eta_u^{dis}} p_{u,t}^{dis} \Delta T, \quad (7)$$

with the initial and terminal conditions:

$$s_{u,0} = s_u^{ini} \quad (8)$$

$$s_{u,T} = s_u^{end}. \quad (9)$$

To prevent charging the BESS from the grid, the following constraint is set:

$$p_{u,t}^{cha} \Delta T \leq e_{u,t}. \quad (10)$$

Moreover, in order to avoid simultaneous charging and discharging, in principle one should impose the constraint $p_{u,t}^{cha} p_{u,t}^{dis} = 0$. However, this constraint can be omitted, being automatically satisfied at the optimum of the optimization problem presented in the following. On the other hand, for the energy exported to and imported from the grid by entity u , the constraint $e_{u,t}^{gri} i_{u,t}^{gri} = 0$ must be enforced. This is done by introducing a binary variable, describing whether the entity exports energy to the grid, or not. In other words:

$$\delta_{u,t} = \begin{cases} 1 & \text{if } e_{u,t}^{gri} > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (11)$$

Using $\delta_{u,t}$ in (11), energy exchanges of entity u with the grid at time t are bounded as follows:

$$0 \leq e_{u,t}^{gri} \leq E_{u,t}^{cap} \Delta T \delta_{u,t} \quad (12)$$

$$0 \leq i_{u,t}^{gri} \leq I_{u,t}^{cap} \Delta T (1 - \delta_{u,t}). \quad (13)$$

Notice that these constraints also impose that there is no simultaneous export to and import from the grid for entity u .

The energy balance for entity u at time t is given by the equation:

$$e_{u,t}^{gri} - i_{u,t}^{gri} - e_{u,t} + i_{u,t} - p_{u,t}^{dis} \Delta T + p_{u,t}^{cha} \Delta T = 0. \quad (14)$$

In order to penalize deviations from the nominal values of energy demand and generation, the terms $|e_{u,t} - E_{u,t}^{nom}|$ and $|i_{u,t} - I_{u,t}^{nom}|$ should be included in the objective function. To linearize these terms, we introduce auxiliary variables $\epsilon_{u,t}^e$ and $\epsilon_{u,t}^i$, and the constraints:

$$e_{u,t} - E_{u,t}^{nom} \leq \epsilon_{u,t}^e \quad (15)$$

$$E_{u,t}^{nom} - e_{u,t} \leq \epsilon_{u,t}^e \quad (16)$$

$$i_{u,t} - I_{u,t}^{nom} \leq \epsilon_{u,t}^i \quad (17)$$

$$I_{u,t}^{nom} - i_{u,t} \leq \epsilon_{u,t}^i. \quad (18)$$

Finally, the virtual self-consumption at time t is expressed as:

$$\min \left\{ \sum_{u \in \mathcal{U}} e_{u,t}^{gri}, \sum_{u \in \mathcal{U}} i_{u,t}^{gri} \right\}. \quad (19)$$

To linearize also these terms in the objective function, we introduce the auxiliary variables ζ_t , and the constraints:

$$\zeta_t \leq \sum_{u \in \mathcal{U}} e_{u,t}^{gri} \quad (20)$$

$$\zeta_t \leq \sum_{u \in \mathcal{U}} i_{u,t}^{gri}. \quad (21)$$

The objective function J , which represents the profit of the community over the time horizon $\mathcal{T}$, including the incentive for the virtual self-consumption, is formulated as follows:

$$\begin{aligned} J = & \sum_{u \in \mathcal{U}} \sum_{t \in \mathcal{T}} [\pi_{u,t}^{egr} e_{u,t}^{gri} - \pi_{u,t}^{igr} i_{u,t}^{gri} \\ & - \pi_{u,t}^{reg} e_{u,t}^e - \pi_{u,t}^{fle} i_{u,t}^i \\ & - \pi_{u,t}^{sto} \Delta T (\eta_u^{cha} p_{u,t}^{cha} + \frac{1}{\eta_u^{dis}} p_{u,t}^{dis})] \\ & + \sum_{t \in \mathcal{T}} \pi_t^{inc} \zeta_t. \end{aligned} \quad (22)$$

This is to be maximized under the constraints (1)-(10), (12)-(18), and (20)-(21), for all $u \in \mathcal{U}$, $u' \in \mathcal{D}$, and $t \in \mathcal{T}$. The resulting problem is a mixed integer linear program (MILP), which can be solved using state-of-the-art solvers, provided that the total number of binary variables is affordable.

At the optimum of the optimization problem, it holds that

$$\zeta_t = \min \left\{ \sum_{u \in \mathcal{U}} e_{u,t}^{gri}, \sum_{u \in \mathcal{U}} i_{u,t}^{gri} \right\} \quad (23)$$

$$\epsilon_{u,t}^e = |e_{u,t} - E_{u,t}^{nom}| \quad (24)$$

$$\epsilon_{u,t}^i = |i_{u,t} - I_{u,t}^{nom}|. \quad (25)$$

B. CM formulation

By applying the technique in [15, Sect. 7.1], the MILP of the previous section can be equivalently rewritten as a continuous optimization problem, in the form of the maximization of a quadratic function with linear constraints. This is done by relaxing the binary variables $\delta_{u,t}$ to be continuous variables over the interval $[0, 1]$, i.e.:

$$0 \leq \delta_{u,t} \leq 1, \quad (26)$$

and adding a suitable regularization term to the objective function (22):

$$J + \gamma \sum_{u \in \mathcal{U}} \sum_{t \in \mathcal{T}} (\delta_{u,t} - \frac{1}{2})^2, \quad (27)$$

where $\gamma > 0$ is a regularization parameter. The resulting problem falls into the category of convex maximization (CM), and its solution can be tackled using the techniques described in [15]. In the numerical results of Section IV, we will show examples where the MILP cannot be solved due to the prohibitive number of binary variables, while the equivalent CM problem is solved in reasonable time.

It turns out that, if γ is chosen sufficiently large, the variables $\delta_{u,t}$ at the optimum of the CM problem are pushed to one of the two boundaries of (26), and therefore they turn out to be binary. In such a situation, the value of J at the optimum of the CM problem coincides with the value at the optimum of the MILP.

III. REDISTRIBUTION STRATEGY

In this section, we introduce a simple redistribution strategy of the incentive for the virtual self-consumption, among the entities of the community. This strategy is characterized by the following properties:

- If an entity has a profit in the community worse than acting individually, it receives an initial compensation which fills the gap.
- The remaining portion of the incentive is divided among the entities proportionally to their contribution to the community self-consumption.

In this way, it is guaranteed that each entity is never penalized by joining the community, compared with acting individually.

The optimization problems formulated in the previous section, take the incentive for the virtual self-consumption explicitly into account in the optimization. This is called the coordinated case (CC), because the incentive couples the behaviors of the entities in view of the maximization of the profit of the community. Alternatively, each entity could optimize its operation independently (uncoordinated case, CU). In that case, the incentive can be computed only a posteriori. We introduce the following quantities:

- J_u^{CU} , J_u^{CC} : profits of entity u in the uncoordinated and coordinated cases;
- I^{CU} , I^{CC} : incentives received by the community in the uncoordinated and coordinated cases.

First, we notice that:

$$J_u^{CC} \leq J_u^{CU}, \quad (28)$$

because in the uncoordinated case each entity maximizes its own profit. Moreover, it holds that:

$$\sum_{u \in \mathcal{U}} J_u^{CU} \leq \sum_{u \in \mathcal{U}} J_u^{CU} + I^{CU} \leq \sum_{u \in \mathcal{U}} J_u^{CC} + I^{CC}. \quad (29)$$

The right-hand side of (29) corresponds to the optimal value of the problems formulated in Section II, which represents the maximum profit achievable including the incentive for the virtual self-consumption. From (28) and (29) we get:

$$I^{CC} \geq \sum_{u \in \mathcal{U}} (J_u^{CU} - J_u^{CC}) \geq 0. \quad (30)$$

These inequalities imply that the incentive I^{CC} can be split into two parts:

$$I^{CC} = I_{comp}^{CC} + I_{red}^{CC}, \quad (31)$$

where I_{comp}^{CC} is equal to $\sum_{u \in \mathcal{U}} (J_u^{CU} - J_u^{CC})$, i.e., it is used to compensate the entities penalized by the coordinated optimization in favor of the profit of the community. The remaining portion $I_{red}^{CC} \geq 0$ can then be divided among the entities proportionally to their contribution to the virtual self-consumption. This shows formally that each entity can be guaranteed to be never penalized by joining the community.

IV. NUMERICAL RESULTS

In this section we first present a toy example which illustrates the redistribution strategy of Section III. Then, we describe a real-world application with 50 entities and different fractions of prosumers.

A. Toy example

We consider a simple community composed of $U = 2$ entities. Entity 1 is a consumer, while entity 2 is a prosumer with a photovoltaic (PV) plant and a BESS. For entity 2 we consider three scenarios.

- Scenario #1: PV plant 4.4 kWp, BESS 0 kWh
- Scenario #2: PV plant 4.4 kWp, BESS 5 kWh
- Scenario #3: PV plant 4.4 kWp, BESS 10 kWh.

For both entities we have real consumption data covering the year 2020 with sampling time $\Delta_T = 1$ hour. The total yearly consumption is 2401 kWh for entity 1, and 3454 kWh for entity 2. We also have PV generation data for entity 2, the total yearly production being 5966 kWh. For both entities, the consumption and generation profiles are considered as the nominal profiles. No deviations from the nominal profiles are allowed. For the BESS of entity 2, the maximum charging and discharging power are set numerically equal to the maximum battery capacity. The charging and discharging efficiencies are chosen equal to 0.95. In the simulations, the initial and terminal battery capacity is set equal to 0. The unitary cost for using BESS is 0.01 €/kWh. Purchase prices are those defined for the *Servizio di Maggior Tutela* by ARERA [16], while selling prices are those defined for the *Ritiro Dedicato* by GSE [17]. Both price data sets refer to year 2020. The unit incentive for virtual self-consumption is equal to 0.12 €/kWh for all time periods (the value currently set in Italy).

Table II shows the aggregated results over one year for entity 2 in the three scenarios. Negative values represents costs, while positive values represent revenues. We denote by $I_{comp,2}^{CC}$ the portion of the incentive I^{CC} assigned for compensation to entity 2. Indeed, $I_{comp,2}^{CC} = J_2^{CU} - J_2^{CC}$. Notice that $I_{comp,1}^{CC} = 0$ for entity 1, which is a consumer with no flexibility, and therefore incurs the same costs in the uncoordinated and coordinated case. Then, $I_{red,2}^{CC}$ is the portion of the remainder of I^{CC} , which is redistributed to entity 2. In this simple case, $I_{red,2}^{CC} = (I^{CC} - I_{comp,2}^{CC})/2$. Clearly, $I_{red,1}^{CC} = I_{red,2}^{CC}$ is the quantity redistributed to entity 1.

It can be noticed in Table II that there is no compensation for entity 2 in Scenario #1, since it incurs the same costs in the uncoordinated and coordinated case. This is because entity 2 is a prosumer with no flexibility. On the other hand, the battery provides flexibility to entity 2 in Scenarios #2 and 3. This flexibility is exploited by the optimization to enhance the overall benefits for the community, but this comes

TABLE II
RESULTS OF THE TOY EXAMPLE [€]

	Scenario #1	Scenario #2	Scenario #3
J_2^{CU}	-148.58	-61.50	-54.98
J_2^{CC}	-148.58	-107.71	-95.78
I^{CC}	70.00	141.66	167.22
$I_{comp,2}^{CC}$	0.00	46.21	40.80
$I_{red,2}^{CC}$	35.00	47.72	63.21
$J_2^{CC,R}$	-113.58	-13.77	8.23

at the cost of a worse individual performance for entity 2 compared with acting outside the community. This situation for entity 2 is reversed after the redistribution, thus showing the importance of a fair redistribution to make remaining within the community the winning choice for each entity.

B. Real-world example

Real demand profiles of 50 consumers with sampling time $\Delta_T = 1$ hour are considered. Data cover the period from April 7, 2008 to October 23, 2008, for a total amount of $D = 200$ days. A normalized profile of PV generation is also available for the same period [18]. These data are used to simulate an energy community with 50 members, different fractions of prosumers (entities with loads, generators and BESS) ranging from 0% to 50%, and different BESS sizes. Since $50 \div 100$ customers are typically connected downstream secondary substations [19], the proposed example could represent a scenario where most of the customers of a low voltage network are joining the energy community. The nominal generation profile $E_{u,t}^{nom}$ for entities u that are prosumers, is obtained by scaling the normalized PV generation profile by the nominal power. We also assume that curtailment is not allowed, thus imposing $E_{u,t}^{max} = E_{u,t}^{nom} = E_{u,t}^{min}$. For the consumers, we do not consider load flexibility, enforcing the constraint $I_{u,t}^{max} = I_{u,t}^{nom} = I_{u,t}^{min}$, where $I_{u,t}^{nom}$ is the real demand profile of entity u .

For the batteries, the minimum capacity is set to $S_u^{min} = 0$, allowing for full discharge, while the maximum capacity S_u^{max} changes in different simulations settings. We consider three different levels for S_u^{max} : no BESS, corresponding to $S_u^{max} = 0$, which means that prosumers are not equipped with batteries; value of S_u^{max} numerically equal to 1.15 and 1.3 times the installed PV power. Charging and discharging efficiencies (η_u^{cha} , η_u^{dis}) are both equal to 0.95. Moreover, the maximum charging and discharging power are set to $P_u^{cha} = P_u^{dis} = S_u^{max} / \Delta_T$. The initial condition of the BESS for day d is equal to the terminal condition of day $d - 1$. For all the prosumers, the batteries cannot be charged with energy coming from the grid, in order to avoid arbitrage, while the values E_u^{cap} and I_u^{cap} are chosen greater than the nominal PV power and the contracted power of each entity.

The prices for energy exported to the grid are those defined for the *Ritiro Dedicato* by GSE [17], while prices for energy imported from the grid are those defined for the *Servizio di*

Maggior Tutela by ARERA [16]. The considered prices range from April to October 2020. To maintain consistency with the load consumption data, price profiles are reconstructed using calendar days from 2008 and price validity periods (day and night hours, workdays and holidays, etc.). According to the current Italian regulation, the unit incentive for the virtual self-consumption is constant at 0.12 €/kWh for all time periods. Additionally, the unitary cost for using BESS is 0.01 €/kWh.

The optimization problems described in Section II are solved for all the $D = 200$ days of the data set. Since $\Delta_T = 1$ hour, one day consists of $T = 24$ time steps. In the literature related to Italian energy communities, the optimization is usually based on mixed-integer formulations [11]–[13]. It is well known that mixed integer problems suffer computational issues, as the number of binary variables grows. This is experienced in the application of this section, where the MILP of Section II-A already cannot be solved when relatively few prosumers, namely 15 out of 50 entities, are present. Conversely, the results presented in Tables III and IV are obtained by solving the CM model described in Section II-B. This model does not show particular computational issues also when applied with 25 prosumers. In all the simulations performed, the value of the regularization coefficient γ is set equal to 10.

Tables III and IV report the average net daily costs for the community in both the coordinated and uncoordinated case. Several fractions of prosumers (with load, PV plant and BESS) are simulated, as well as different values of maximum battery capacity, as described above. The results obtained suggest that the coordination among members, which corresponds to sharing information about demand, generation, and battery levels, leads benefits to the entire community. This is mainly related to the incentive obtained for the virtual self-consumption, which is greater in the coordinated case. For the sake of a fair comparison, we stress that this analysis focuses only on operating costs and revenues, and does not include investment costs for the hardware required by the coordination.

V. CONCLUSIONS

This paper addressed the optimal operation of an energy community, which receives an incentive for the energy virtually shared by its members. The considered problem was formulated as a MILP, with the objective to maximize the profit of the community (which includes the incentive for the

TABLE III
AVERAGE NET DAILY COSTS, COORDINATED CASE [€]

	NO BESS	+15%	+30%
0%	544.24	544.24	544.24
10%	483.76	483.87	483.89
20%	421.20	420.30	420.28
30%	354.41	348.26	347.86
40%	309.16	295.08	293.92
50%	267.11	229.71	226.84

TABLE IV
AVERAGE NET DAILY COSTS, UNCOORDINATED CASE [€]

	NO BESS	+15%	+30%
0%	544.24	544.24	544.24
10%	483.76	487.28	487.50
20%	421.20	429.68	430.39
30%	354.41	364.22	364.94
40%	309.16	318.19	318.99
50%	267.11	262.01	261.45

virtual self-consumption) over a given time horizon. Solving the problem provides the BESS charging/discharging policies, and the set points of controllable loads and generators. To cope with the computational issues of MILP formulations when the number of binary variables increases, an equivalent formulation in the form of convex maximization was derived, involving only continuous variables. An example of realistic size was presented, where the latter problem could be solved in reasonable time, while solving the MILP was not computationally affordable. Moreover, a simple strategy to fairly redistribute the community benefits among the members was discussed. A toy example was presented to highlight the importance of such a task when a centralized optimization scheme is adopted. Future work is concerned with introducing uncertainty on demand and generation, through a stochastic formulation of the problem at hand. Moreover, participation of the community in the provision of different services, such as demand response and charging of electric vehicles, will be considered.

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